



MALDIVES POLYTECHNIC

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Course	ME03ARM	Semester	2026 / Sem I

Assignment Details:

Subject / Module	Automotive Engine Practice		
Lecturer / Tutor	Ibrahim Siraj	Subject code	AEP01
Assignment Name	Engine Disassembly and Assembly		
Due date	21 April 2026	Date & Time submitted	21 April 2026

Student declaration:

I <u>Hussain Shareef</u> Hereby declare this assignment is my own work. And I also declare that the assignment has not been submitted for assessment to Maldives Polytechnic or elsewhere before in this module or other. Furthermore I declare that I have not included any plagiarized or colluded contents. Sign:.....
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A male student with a beard, wearing a dark blue polo shirt, is focused on working with a lathe machine in a workshop. The background is slightly blurred, showing other equipment and a bright, airy environment.

MALDIVES POLYTECHNIC

Engine Disassembly and Assembly Report

Nissan QG15DE practical
assignment for Automotive
Engine Practice.

Student: Hussain Shareef

Course: ME03ARM - Certificate III in
Automotive Repair & Maintenance

Module: AEP01 - Automotive Engine
Practice

Engine: Nissan QG15DE

Date: 21 April 2026

1. Introduction

This practical test was carried out to understand the correct procedure for disassembling, inspecting, measuring, and assembling an internal combustion engine. The engine used for this practical was the Nissan QG15DE petrol engine.



Figure 1. Nissan QG15DE engine mounted on the stand before practical disassembly work began. This stage was used to identify external components and prepare the engine for teardown.

The main objective of this practical was to identify major engine components, remove them in the correct sequence, inspect them for wear or damage, take important measurements, and reassemble the engine using correct workshop procedures. During the activity, safety, cleanliness, correct tool usage, and manufacturer specifications were followed as important requirements.

Objective

The objectives of this practical were:

- To prepare a safe and clean workshop area for engine work.
- To identify the main external and internal engine components.
- To disassemble the engine step by step.
- To inspect engine components for cracks, scoring, carbon buildup, and excessive wear.
- To measure important clearances and compare them with specifications.
- To assemble the engine correctly using lubrication, correct alignment, and correct tightening procedures.
- To understand the importance of timing marks, torque specifications, and final engine checks.

2. Engine Specifications

The engine used in this practical was the Nissan QG15DE.

Item	Specification
Engine model	Nissan QG15DE

Item	Specification
Engine type	Petrol / gasoline
Cylinder arrangement	4-cylinder, in-line
Valve arrangement	DOHC
Displacement	1,497 cm ³
Bore x stroke	73.6 mm x 88.0 mm
Firing order	1-3-4-2
Compression ratio	9.9:1
Number of main bearings	5
Piston rings	2 compression rings and 1 oil ring
Standard spark plug	NGK BKR5E-11
Spark plug gap	1.0 - 1.1 mm
Standard compression pressure	1,372 kPa / 198.9 psi at 350 rpm
Minimum compression pressure	1,176 kPa / 170.5 psi
Maximum difference between cylinders	98 kPa / 14 psi

Additional details from the practical engine:

Item	Practical Engine Detail
Make	Nissan
Model	QG15DE
Year	1999
Serial number	Not confirmed - to be checked from engine block stamping if visible
Vehicle/model source	_____

3. Workspace Preparation

Before starting the disassembly process, the workspace was prepared properly. A clean and organized work area is important because engine components are precision parts and can be damaged by dirt, incorrect storage, or mixing up parts.

The engine was placed securely on an engine stand before work started. The stand allowed safe access to the top, bottom, front, and rear sections of the engine. Tools were arranged before starting, and containers were prepared for bolts, nuts, washers, and small components.

The following preparation steps were followed:

- The work area was cleaned and cleared of unnecessary items.
 - The engine stand was checked to confirm it was stable and secure.
 - Required tools were arranged before starting the work.
 - Containers or trays were prepared to keep removed parts in order.
 - Components such as pistons, connecting rods, bearing caps, and bolts were marked or arranged according to cylinder number.
 - Waste containers were prepared for used oil and coolant.
-

4. Safety and PPE

Safety precautions were followed throughout the practical. Engine work involves heavy parts, sharp edges, hot fluids, chemical cleaners, and hand tools, so Occupational Health and Safety requirements must be followed.

PPE Used

- Safety shoes
- Workshop overalls
- Safety gloves
- Safety goggles or eye protection
- Face mask when cleaning carbon or gasket material

Safety Precautions

- The engine was supported securely before work started.
 - The work area was kept clean and dry to prevent slipping.
 - Oil and coolant were drained into suitable containers.
 - Fuel lines and electrical connectors were handled carefully.
 - Correct tools were used to avoid damaging bolts or components.
 - Heavy components were lifted carefully with assistance when required.
 - Removed parts were stored safely and labelled where necessary.
-

5. Pre-Disassembly Procedures

Before disassembling the engine, external checks and preparation were completed.

Fluid Drainage

The engine oil was drained by removing the oil drain plug and allowing the oil to flow into a suitable waste oil container. The coolant was drained from the cooling system into a separate container. Oil and coolant were not allowed to spill onto the floor because this can create a slipping hazard and cause environmental contamination.

External Component Removal

The following external components were removed before internal engine disassembly:

- Battery terminals were disconnected if the engine was still installed in the vehicle.
- Air intake components were removed.
- Intake manifold was removed or disconnected.
- Exhaust manifold was removed.
- Alternator was removed.
- Water pump and related cooling components were removed as required.
- Electrical connectors were disconnected carefully.
- Fuel lines and vacuum hoses were disconnected and identified.
- Drive belts were removed.
- Spark plugs and ignition coils were removed.

During removal, bolts and nuts were kept in order to make assembly easier.

6. Engine Disassembly Procedure

6.1 Cylinder Head Removal

The valve cover was removed first to expose the camshafts and valve train area. The No. 1 piston was positioned at Top Dead Center on the compression stroke before timing components were removed.

The timing marks were checked and recorded before the timing chain was removed. This is important because incorrect timing during assembly can cause poor engine operation or valve damage.

The cylinder head bolts were loosened gradually in the correct sequence. The bolts were loosened in two or three stages to prevent cylinder head warping. Removing the cylinder head bolts in the wrong order can cause distortion or cracking.

Main steps:

1. Remove valve cover.
2. Remove ignition coils and spark plugs.
3. Set No. 1 piston at TDC on compression stroke.
4. Mark or record timing chain alignment marks.
5. Remove timing chain components carefully.
6. Loosen cylinder head bolts gradually in the recommended order.
7. Remove the cylinder head from the engine block.
8. Place the cylinder head on a clean surface for inspection.

6.2 Timing System Removal

The QG15DE uses a timing chain system. During removal, the timing marks on the chain and sprockets were observed and recorded. The chain tensioner and guides were removed carefully.



Figure 2. Timing chain, camshaft sprockets, and front timing components exposed before removal. Timing alignment was checked before the chain was removed.

Important points:

- Timing marks must be aligned before removal.
- Camshaft and crankshaft should not be rotated separately after removing the chain.
- Chain guides and tensioner must be inspected for wear.
- Timing components should be stored in order.

6.3 Bottom End Teardown

After the cylinder head and timing components were removed, the bottom end was disassembled.



Figure 3. Pistons and connecting rods arranged after removal. Keeping these parts in order prevents mixing components that have worn together during engine operation.

Main steps:

1. Remove the oil pan.
2. Remove the oil strainer and oil pump if required.

3. Remove connecting rod caps.
4. Push piston and connecting rod assemblies out carefully.
5. Number each piston and connecting rod according to its cylinder.
6. Remove main bearing caps.
7. Remove the crankshaft.
8. Remove main bearings and thrust bearings.
9. Store bearings and caps in correct order.

It is important that connecting rods, pistons, bearing caps, and bearings are not mixed up because they wear together during engine operation.

7. Inspection and Measurement

After disassembly, all components were cleaned and inspected. Carbon buildup, old gasket material, and oil deposits were removed carefully. A scraper and suitable cleaning tools were used, but sealing surfaces were not scratched or damaged.

Visual Inspection

The following checks were carried out:

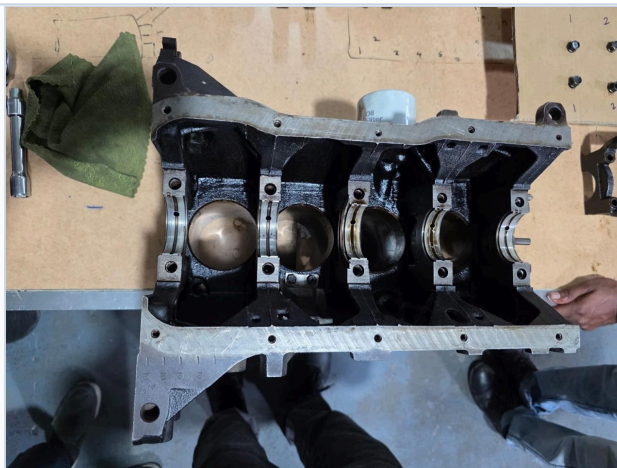


Figure 4. Cylinder block lower section and main bearing saddles after crankshaft removal. Bearing surfaces were inspected for scoring, abnormal wear, and cleanliness.

- Cylinder head checked for cracks, corrosion, and warping.
- Cylinder block checked for cracks, scoring, and damaged threads.
- Cylinder walls checked for scoring, wear, and seizure marks.
- Pistons checked for cracks, carbon buildup, and skirt wear.
- Piston rings checked for wear and damage.
- Crankshaft journals checked for scoring and wear.
- Bearings checked for scratches, copper exposure, and abnormal wear.
- Timing chain, guides, and tensioner checked for wear.

Measurement Specifications

Measurement	QG15DE Specification
Cylinder head flatness standard	Less than 0.03 mm
Cylinder head flatness limit	0.1 mm
Cylinder bore Grade 1	73.600 - 73.610 mm
Cylinder bore Grade 2	73.610 - 73.620 mm
Cylinder bore Grade 3	73.620 - 73.630 mm
Cylinder bore wear limit	0.2 mm
Cylinder bore out-of-round	Less than 0.015 mm
Cylinder bore taper	Less than 0.01 mm
Piston-to-bore clearance	0.015 - 0.035 mm
Top ring end gap	0.20 - 0.30 mm
Second ring end gap	0.32 - 0.47 mm
Oil ring end gap	0.20 - 0.60 mm
Crankshaft end play standard	0.060 - 0.220 mm
Crankshaft end play limit	0.3 mm
Main bearing clearance	0.020 - 0.044 mm
Main bearing clearance limit	0.1 mm
Connecting rod bearing clearance	0.014 - 0.039 mm
Connecting rod bearing clearance limit	0.1 mm
Connecting rod side clearance	0.200 - 0.470 mm
Connecting rod side clearance limit	0.5 mm

Practical Measurement Record

Item Measured	Cylinder/Position	Measured Value	Specification	Result
Cylinder bore	1	73.61 mm	73.600 - 73.630 mm	Pass
Cylinder bore	2	73.62 mm	73.600 - 73.630 mm	Pass
Cylinder bore	3	73.61 mm	73.600 - 73.630 mm	Pass
Cylinder bore	4	73.62 mm	73.600 - 73.630 mm	Pass
Piston-to-bore clearance	-	0.020 mm	0.015 - 0.035 mm	Pass
Top ring end gap	1	0.25 mm	0.20 - 0.30 mm	Pass

Item Measured	Cylinder/Position	Measured Value	Specification	Result
Second ring end gap	1	0.33 mm	0.32 - 0.47 mm	Pass
Oil ring end gap	1	0.50 mm	0.20 - 0.60 mm	Pass
Crankshaft end play	-	0.090 mm	0.060 - 0.220 mm	Pass
Main bearing clearance	-	0.030 mm	0.020 - 0.044 mm	Pass
Connecting rod bearing clearance	-	0.020 mm	0.014 - 0.039 mm	Pass
Connecting rod side clearance	-	0.250 mm	0.200 - 0.470 mm	Pass
Cylinder head flatness	-	0.02 mm	Less than 0.03 mm	Pass

Additional Recorded Measurements from Evidence Photos

Component	Measurement / Position	Recorded Value	Evidence Source
Crankshaft	Main journal reading 1	49.06 mm	See Appendix B: B1. Crankshaft measurement record sheet; B3. Crankshaft journal digital notes; B4. Crankshaft and valve measurement record sheet
Crankshaft	Main journal reading 2	49.48 mm	See Appendix B: B1. Crankshaft measurement record sheet; B3. Crankshaft journal digital notes; B4. Crankshaft and valve measurement record sheet
Crankshaft	Main journal reading 3	49.03 mm	See Appendix B: B1. Crankshaft measurement record sheet; B3. Crankshaft journal digital notes; B4. Crankshaft and valve measurement record sheet
Crankshaft	Main journal reading 4	49.37 mm	See Appendix B: B1. Crankshaft measurement record sheet; B3. Crankshaft journal digital notes; B4. Crankshaft and valve measurement record sheet
Crankshaft	Pin journal readings	39.96 mm, 39.98 mm, and 39.58 mm	See Appendix B: B1. Crankshaft measurement record sheet; B4. Crankshaft and valve measurement record sheet
Camshaft	Cam height readings	40.00 mm	See Appendix B: B2. Intake camshaft measurement record sheet
Camshaft	Front journal readings	27.95 mm and 27.92 mm	See Appendix B: B2. Intake camshaft measurement record sheet
Camshaft	Camshaft journal readings	24.44 mm and 24.45 mm	See Appendix B: B2. Intake camshaft measurement record sheet
Inlet valve	Stem diameter	5.98 mm	See Appendix B: B6. Inlet valve measurement detail; B7. Inlet valve measurement close view
Inlet valve	Face/head diameter	29.90 mm	See Appendix B: B5. Piston pin and valve measurement record sheet; B6. Inlet valve measurement detail; B7. Inlet valve measurement close view

Component	Measurement / Position	Recorded Value	Evidence Source
Piston pin	Diameter readings	40.16 mm and 40.00 mm	See Appendix B: B5. Piston pin and valve measurement record sheet

Measurement Record Evidence

The following handwritten evidence supports the recorded measurement values used in the report.

Evidence No.	Description
B1. Crankshaft measurement record sheet	Crankshaft journal measurement record sheet
B2. Intake camshaft measurement record sheet	Intake camshaft journal and cam height measurement record sheet
B3. Crankshaft journal digital notes	Digital notes showing crankshaft journal measurement positions
B4. Crankshaft and valve measurement record sheet	Crankshaft, pin journal, and valve measurement record sheet
B5. Piston pin and valve measurement record sheet	Piston pin and valve measurement record sheet
B6. Inlet valve measurement detail	Inlet valve measurement detail
B7. Inlet valve measurement close view	Close view of inlet valve measurement detail

8. Engine Assembly Procedure

Before assembly, all parts were cleaned and inspected. Bearing surfaces, cylinder walls, piston rings, crankshaft journals, and bolt threads were lubricated with clean engine oil where required.

8.1 Block Preparation

The cylinder block was cleaned and checked before installation of internal components. Oil passages and coolant passages were checked for dirt or blockage. Old gasket material was removed from sealing surfaces.

Preparation steps:

- Clean cylinder block surface.
- Clean bearing saddles and caps.
- Clean crankshaft journals.
- Apply engine oil to bearing surfaces.
- Install correct main bearings in their original positions.
- Install thrust bearings correctly.

8.2 Crankshaft Installation

The crankshaft was installed into the cylinder block after the main bearings were fitted. Main bearing caps were installed in the correct order and direction.

Important points:

- Main bearing caps must be installed in their original positions.
- Bearing surfaces must be lubricated with clean engine oil.
- Main bearing cap bolts should be tightened gradually.
- The crankshaft must rotate smoothly by hand after tightening.
- Crankshaft end play must be checked after installation.

8.3 Piston and Connecting Rod Installation

Piston rings were installed on the pistons. The ring gaps were positioned correctly around the piston and not aligned together. A piston ring compressor was used to install the piston and connecting rod assemblies into the cylinders.

Important points:

- Pistons and rods must be installed into their correct cylinders.
- The front mark on the piston must face the front of the engine.
- Connecting rod bolts must not scratch the crankshaft journals.
- Piston rings and cylinder walls must be lubricated.
- Connecting rod caps must match their original rods.

8.4 Cylinder Head Installation

A new cylinder head gasket was installed before fitting the cylinder head. The cylinder head bolts were tightened using the manufacturer procedure.

For the QG15DE, the cylinder head bolt procedure is:

Step	Procedure
1	Tighten bolts to 29 Nm
2	Tighten bolts to 59 Nm
3	Loosen bolts completely
4	Tighten bolts again to 29 Nm
5	Turn bolts 50 - 55 degrees clockwise
6	Tighten small front bolts to 6.3 - 8.3 Nm

Important note: Cylinder head bolts require angular tightening. Bolt threads and seat surfaces should be clean and coated with engine oil.

8.5 Connecting Rod Cap Nut Tightening

For the QG15DE, the connecting rod cap nut procedure is:

Step	Procedure
1	Tighten to 13.72 - 15.68 Nm
2	Turn 35 - 40 degrees clockwise using an angle wrench

Alternative if an angle wrench is not available:

Alternative Procedure
Tighten to 23 - 28 Nm

8.6 Timing Alignment

The timing chain was installed by aligning the timing marks on the crankshaft sprocket, camshaft sprockets, and timing chain. The chain tensioner and guides were installed correctly.

Important points:

- No. 1 piston must be at TDC.
- Timing marks must align correctly.
- Camshaft and crankshaft should not be rotated separately before timing is installed.
- Chain tensioner must be correctly released after installation.
- Engine should be rotated by hand to confirm there is no interference.

9. Final Checks and Testing

After assembly, final checks were completed before considering the engine ready.

The following checks were carried out:

- Confirmed all bolts and nuts were tightened.
- Confirmed the crankshaft rotated freely by hand.
- Checked timing marks after rotating the engine.
- Checked that pistons and valves did not contact.
- Checked that all gaskets and seals were correctly installed.
- Checked that oil pan and covers were installed properly.
- Checked that external components were refitted correctly.
- Checked that electrical connectors, hoses, and fuel lines were connected.

If the engine is installed in a vehicle, further checks should include:

- Refill engine oil.
- Refill coolant.
- Check for oil and coolant leaks.
- Start the engine and listen for abnormal noise.
- Check oil pressure warning light.
- Check engine temperature.
- Carry out compression test if required.

10. Tools and Equipment Used

The following tools and equipment were used or required during the practical:

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- Engine stand
 - Socket set and ratchet
 - Spanners
 - Screwdrivers
 - Torque wrench
 - Angle wrench
 - Piston ring compressor
 - Feeler gauge
 - Bore gauge
 - Micrometer
 - Dial gauge
 - Scraper
 - Plastic hammer
 - Oil drain container
 - Coolant drain container
 - Cleaning solvent
 - Engine oil for lubrication
-

11. Photos and Evidence

Selected practical photos are included in the report body, and the full evidence set is provided in Appendix A.

Appendix A Ref.	Description
A1. Engine before disassembly - intake side	Engine before disassembly, intake side view
A2. Engine before disassembly - exhaust side	Engine before disassembly, exhaust side view
A3. Valve cover installed	Valve cover installed with spark plugs removed
A4. Valve cover removed	Valve cover removed and placed separately
A5. Timing chain front view	Timing chain and camshaft sprockets, front view
A6. Timing chain side view	Timing chain and timing components, side view
A7. Timing chain tensioner removal	Timing chain tensioner removal process
A8. Camshafts and timing components laid out	Camshafts, sprockets, chain, guides, and caps arranged in order
A9. Cylinder head and camshafts exposed	Cylinder head top section with camshafts exposed
A10. Intake manifold removal	Intake manifold being removed from the engine
A11. Intake manifold detail	Intake manifold and fuel rail detail
A12. Exhaust manifold installed	Exhaust manifold installed on the engine

Appendix A Ref.	Description
A13. Exhaust manifold removal	Exhaust manifold removal process
A14. Exhaust manifold and catalytic converter removed	Exhaust manifold and catalytic converter removed
A15. Coolant outlet and pump installed	Coolant outlet and pump area installed on the engine
A16. Coolant outlet detail	Coolant outlet detail after removal
A17. Single piston and connecting rod removed	Single piston and connecting rod assembly removed
A18. Pistons and connecting rods arranged	Pistons and connecting rods arranged in order
A19. Crankshaft removed	Crankshaft removed and placed on the workbench
A20. Main bearing saddles and cylinder block	Cylinder block lower section and main bearing saddles
A21. Student performing engine work	Student carrying out practical engine work

12. Conclusion

In conclusion, this practical helped develop an understanding of the construction, disassembly, inspection, measurement, and assembly of the Nissan QG15DE engine. The practical showed the importance of following correct procedures, using suitable tools, keeping parts organized, and applying manufacturer specifications.

The activity also showed that engine assembly requires accuracy and cleanliness. Important components such as the crankshaft, pistons, connecting rods, bearings, cylinder head, and timing chain must be installed correctly. Measurements such as cylinder bore, piston ring gap, crankshaft end play, and bearing clearance are necessary to determine whether parts are serviceable.

The final check confirmed that the engine should rotate freely by hand and that all fasteners must be secure before the engine is started or returned to service.

13. References

- Nissan QG15DE Engine Mechanical Service Manual, Section EM.

Appendix A: Practical Photo Evidence

The following photos show the engine before disassembly, component removal, timing system, internal engine parts, and practical workshop activity.



A1. Engine before disassembly - intake side



A2. Engine before disassembly - exhaust side



A3. Valve cover installed with spark plugs removed



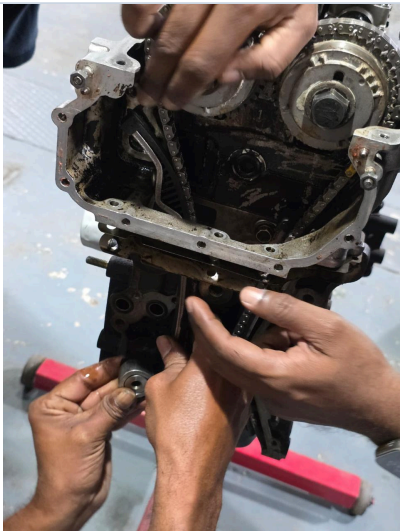
A4. Valve cover removed



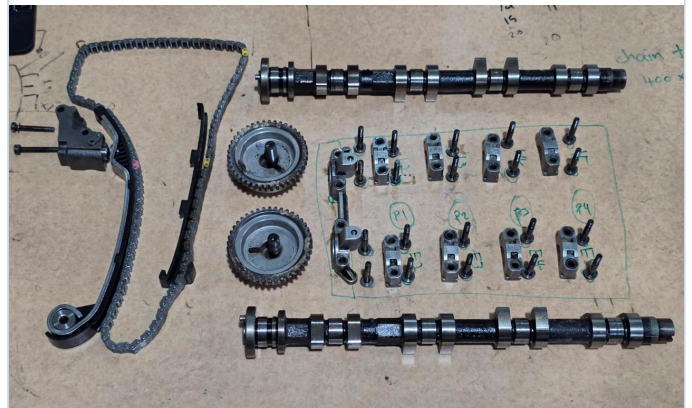
A5. Timing chain front view



A6. Timing chain side view



A7. Timing chain tensioner removal



A8. Camshafts and timing components laid out



A9. Cylinder head and camshafts exposed



A10. Intake manifold removal



A11. Intake manifold detail



A12. Exhaust manifold installed



A13. Exhaust manifold removal



A14. Exhaust manifold and catalytic converter removed



A15. Coolant outlet and pump installed



A16. Coolant outlet detail



A17. Single piston and connecting rod removed



A18. Pistons and connecting rods arranged



A19. Crankshaft removed



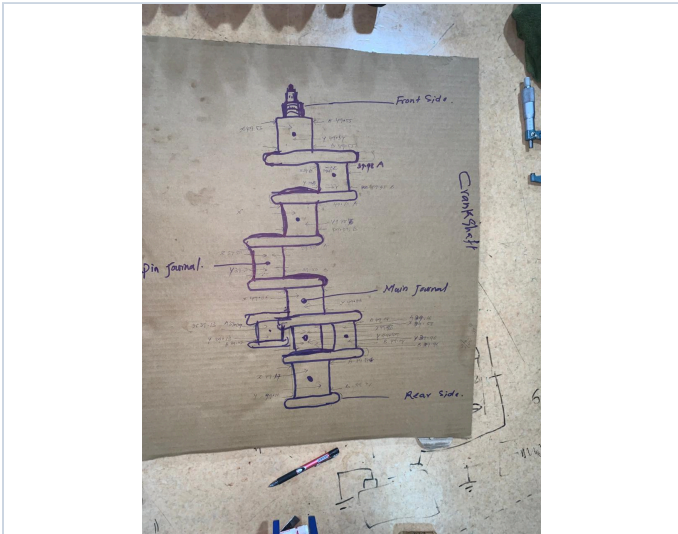
A20. Main bearing saddles and cylinder block



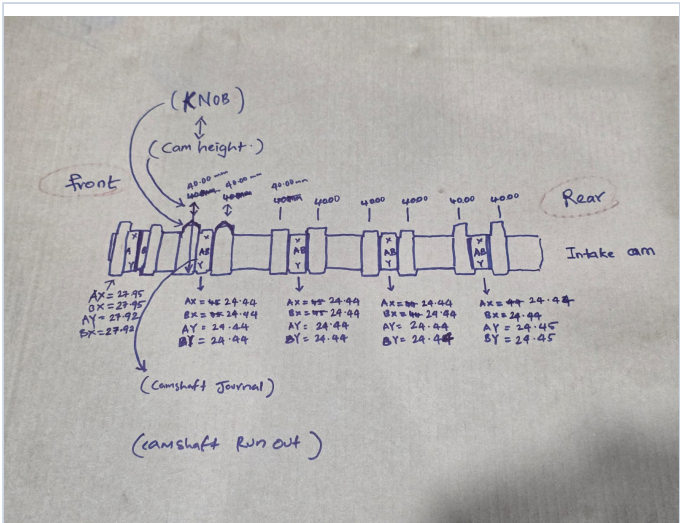
A21. Student performing engine work

Appendix B: Measurement Record Evidence

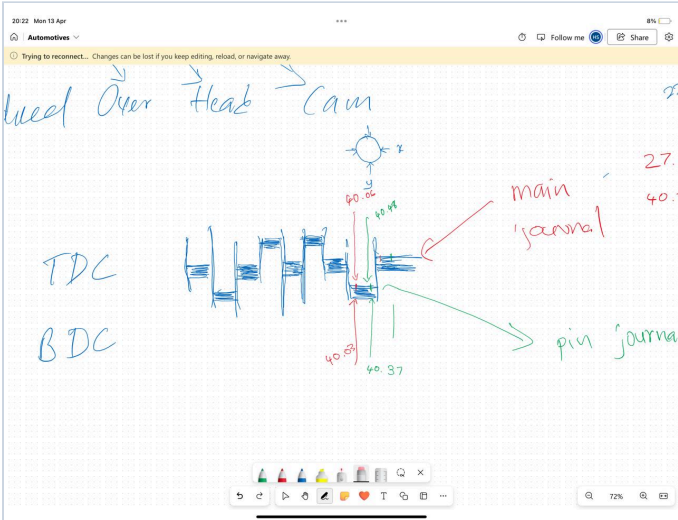
The following photos show handwritten workshop measurement diagrams and record sheets used during the practical.



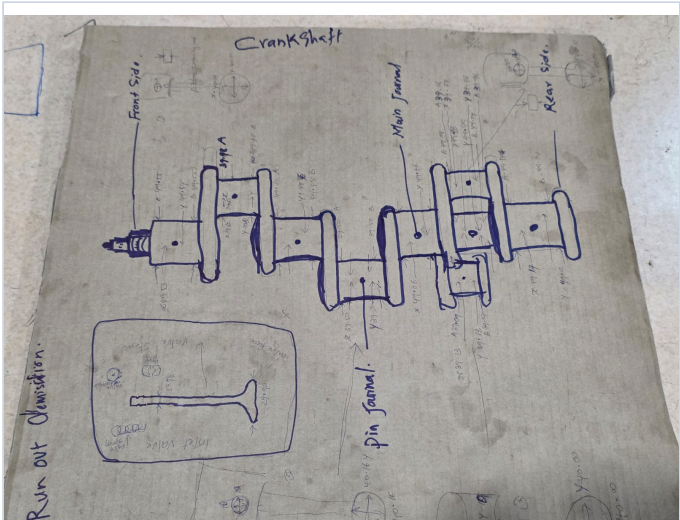
B1. Crankshaft measurement record sheet



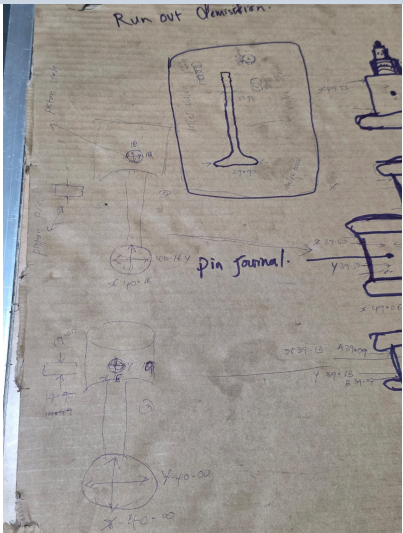
B2. Intake camshaft measurement record sheet



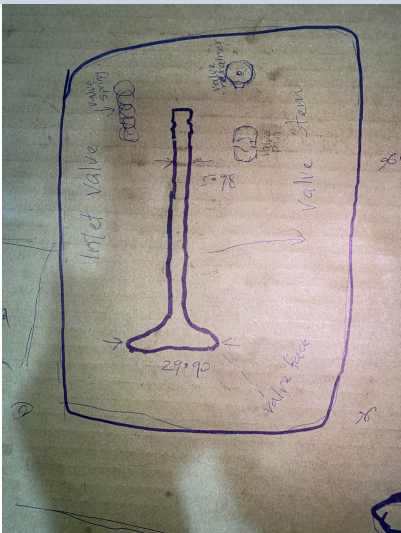
B3. Crankshaft journal digital notes



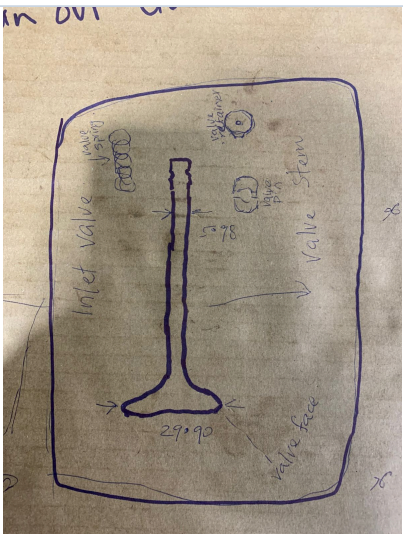
B4. Crankshaft and valve measurement record sheet



B5. Piston pin and valve measurement record sheet



B6. Inlet valve measurement detail



B7. Inlet valve measurement close view